

ME597: Machine Learning (ML)

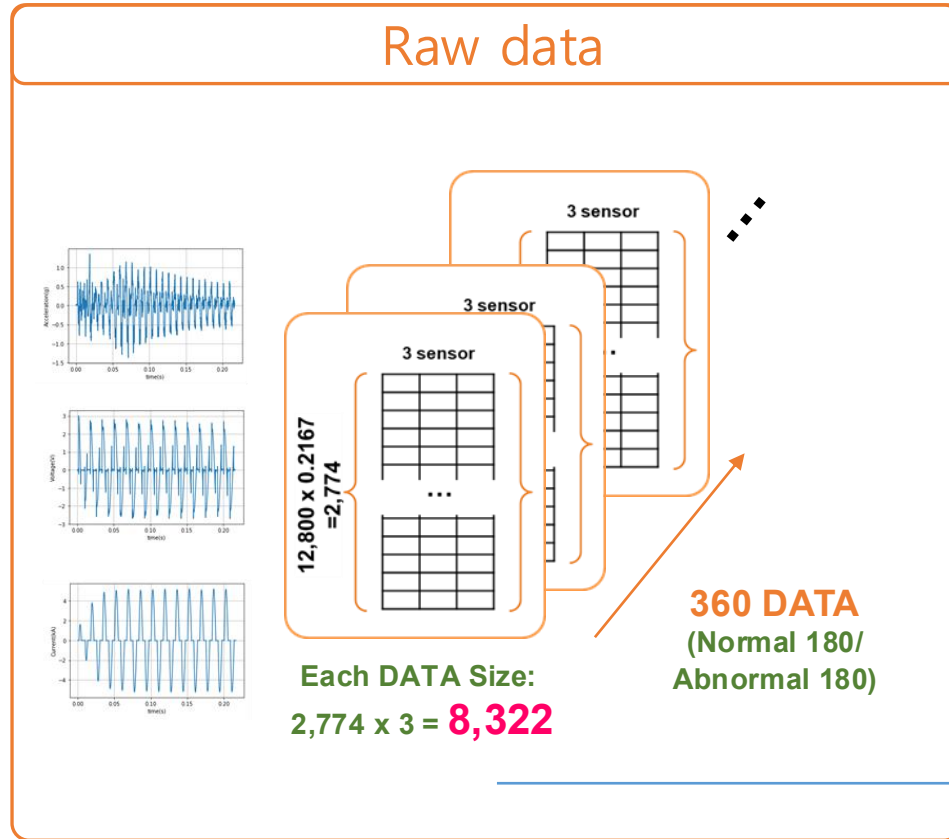
Tutorial 4

Week 3

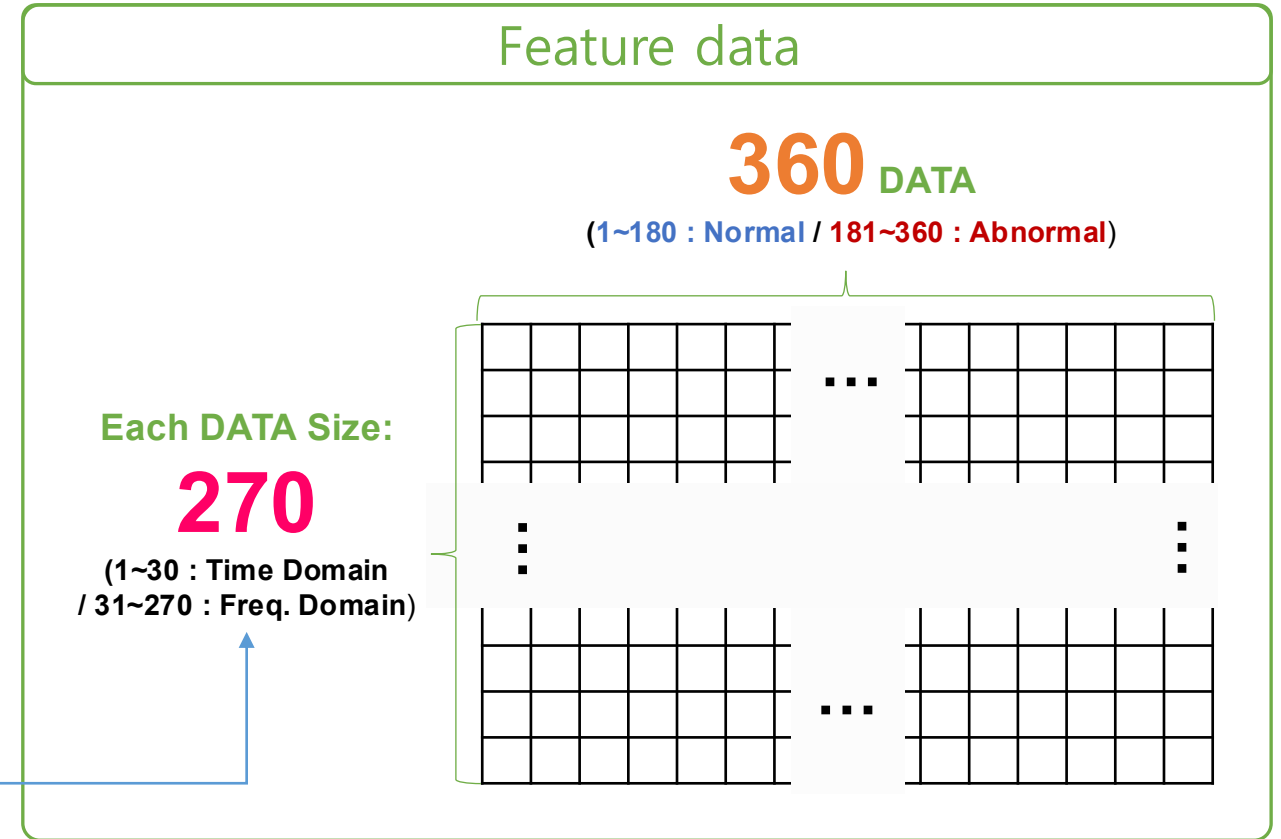
School of Mechanical Engineering
Purdue University

Dr. Martin Jun

Feature Extraction

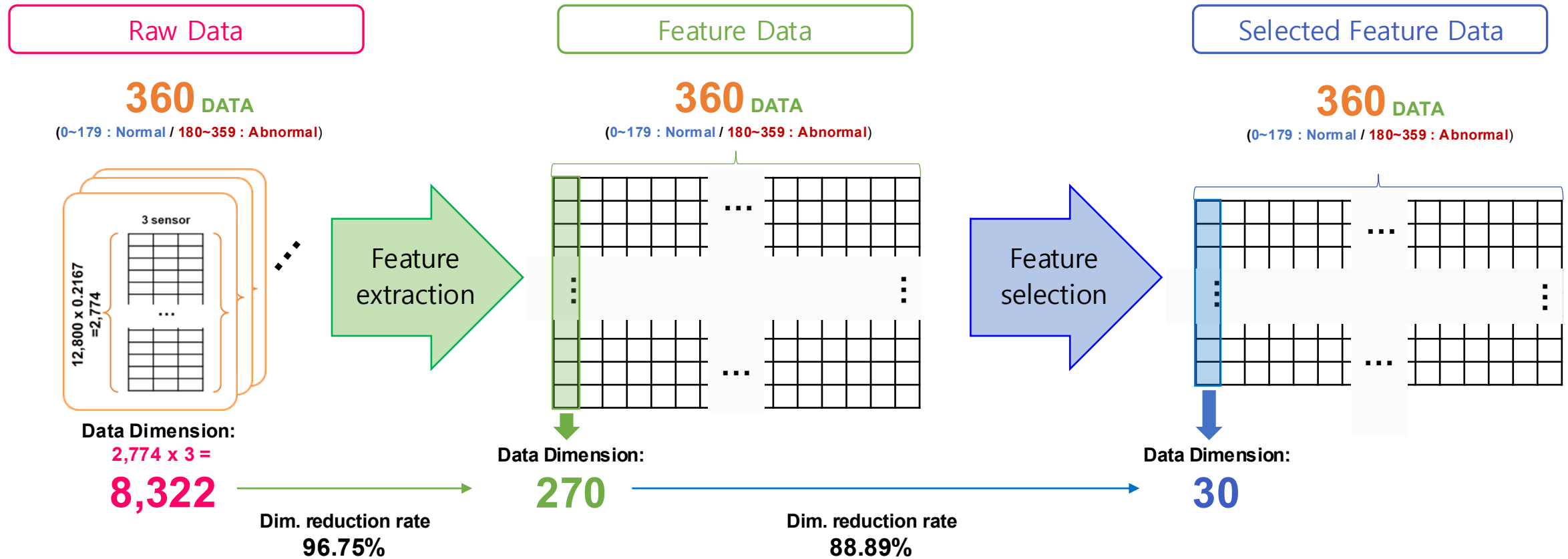


360 Data files
(.csv)

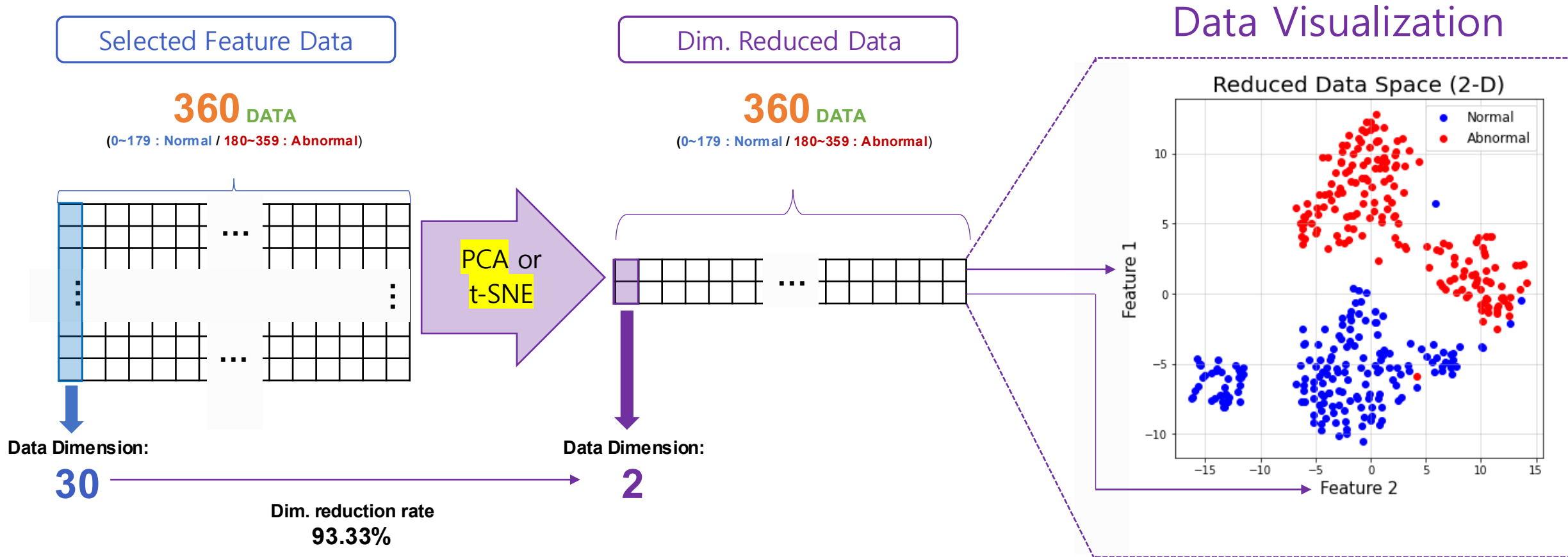


1 Data files
(.csv)

Feature Extraction



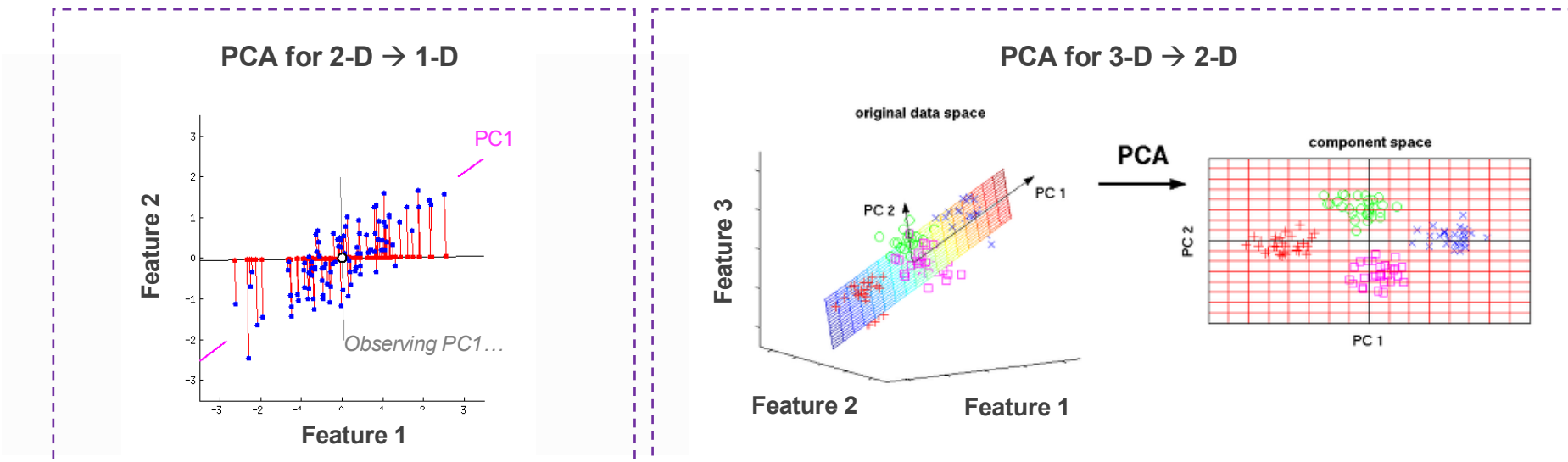
Data Visualization (2-D space)



Principal Component Analysis (PCA)

PCA : Principal Component Analysis

- Most famous (unsupervised) data dimensionality reduction method
 - Project data (feature) onto a lower dimensional linear space such that the variance of the projected data is maximized



The axis with the largest variance of the projected features is selected as the 1st principal component (PC) by rotating around the central point of the feature data.

→ Search the 2nd PC in a direction **perpendicular to the 1st PC**

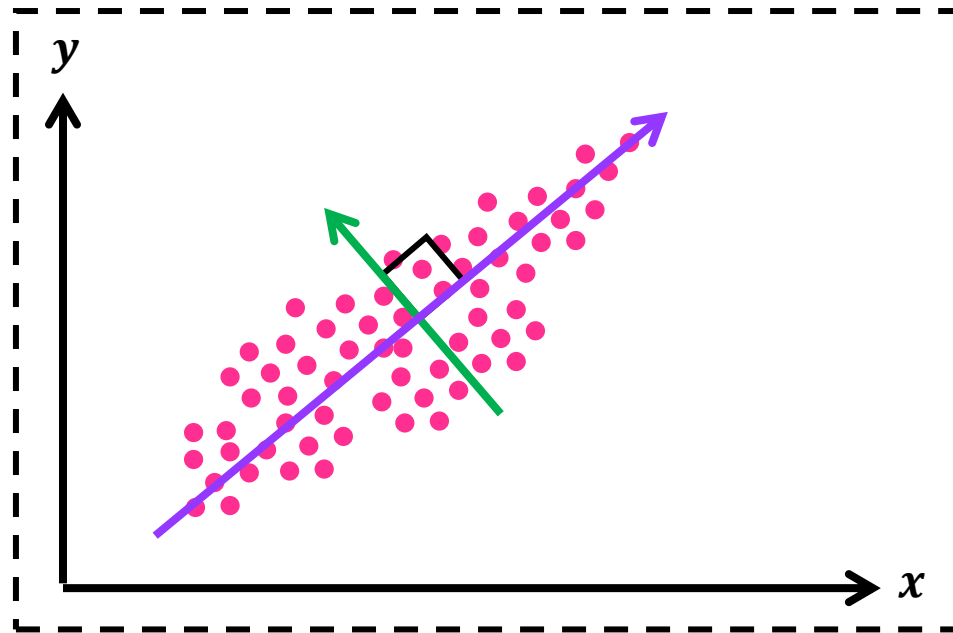
→ Search the 3rd PC in a direction perpendicular to the 2nd PC

...

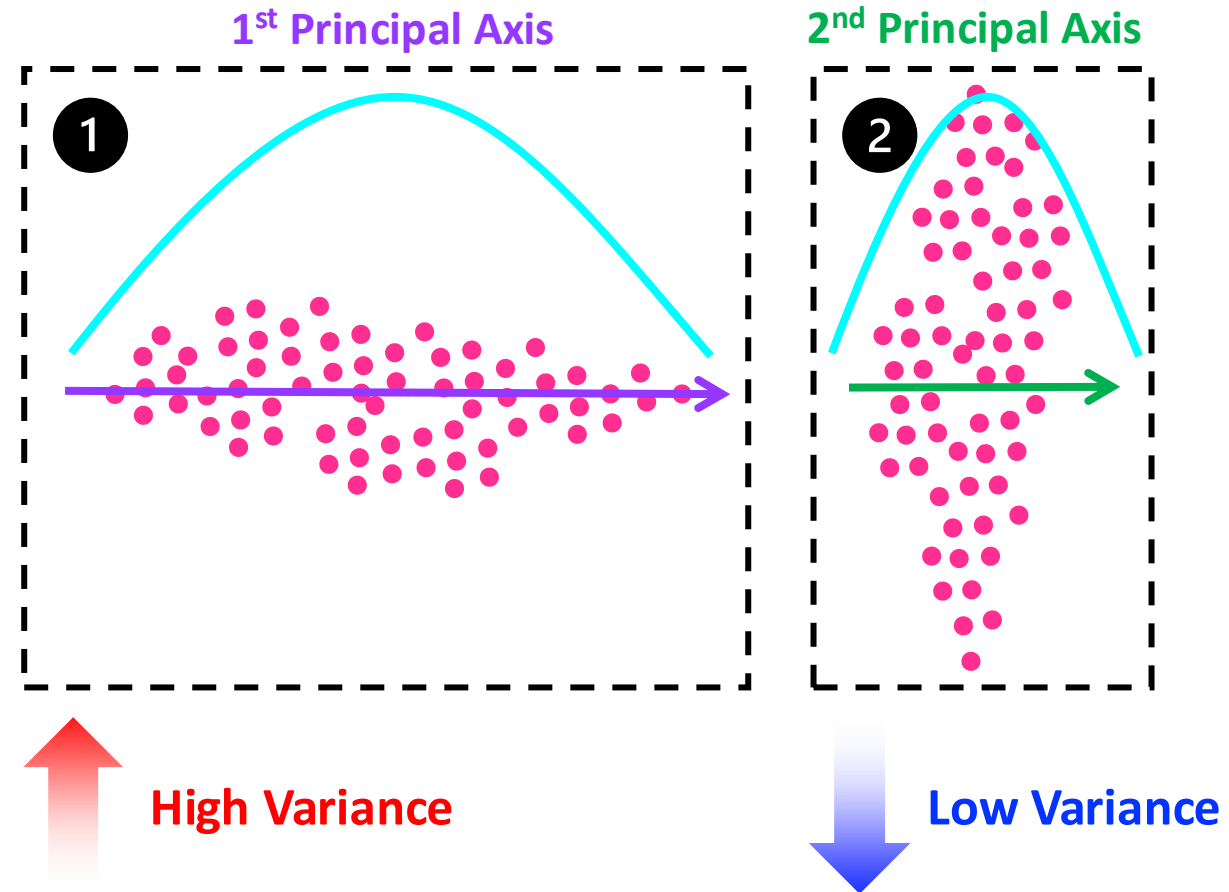
Principal Component Analysis (PCA)

Lower Dimensional Linear Space? + Maximizing Variance?

Simplified 2D Example



Dimensional
Reduction



Principal Component Analysis (PCA)

(Pre-processing step)

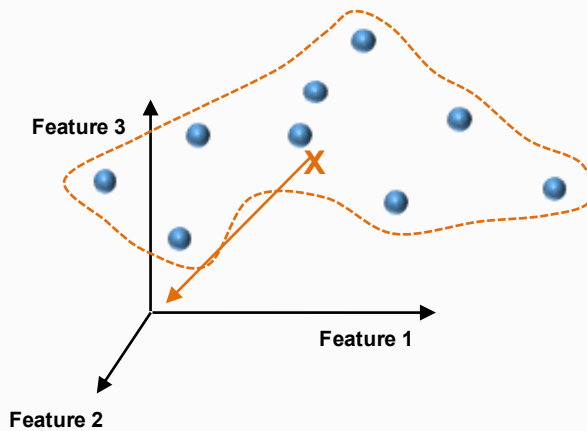
Step 1

"Zero mean"

Replace x^i with $(x^i - \mu)$

$$\text{where, } \mu = \frac{1}{m} \sum_{i=1}^m x^i$$

Move the central point of features to the origin



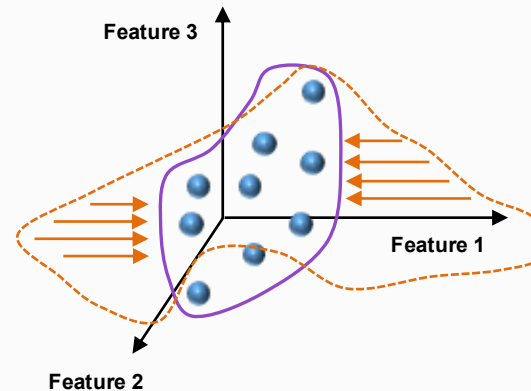
Step 2

"Unit Variance"

Replace $(x^i - \mu)$ with $\frac{x^i - \mu}{\sigma}$

$$\text{where, } \sigma^2 = \frac{1}{m} \sum_{i=1}^m (x^i)^2$$

Feature data $\rightarrow$ Standardized data

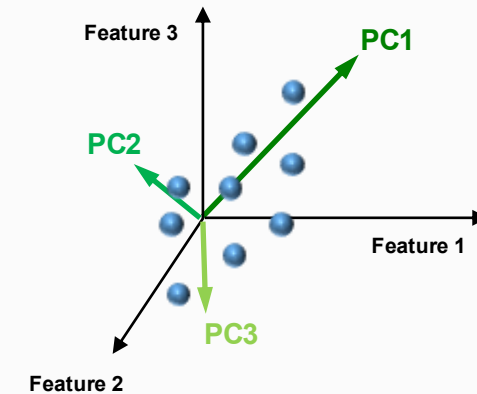


Step 3

"PCA"

Optimization problem with a constraint:

- Find u that maximize $u^T \sum u$
- u : Principal Component (PC)
= coefficient matrix (*coeff*)
- *Score* : **Standardized data** $\cdot$ *coeff*



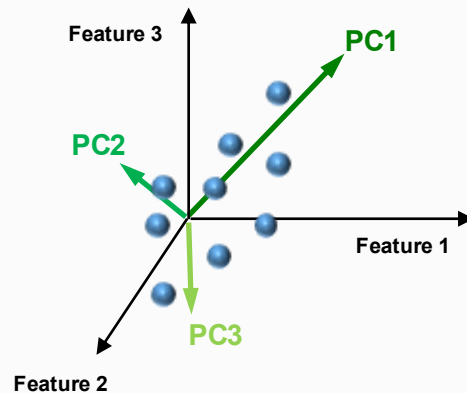
Principal Component Analysis (PCA)

Step 3

"PCA"

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- Find u that maximize $u^T \sum u$
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Covariance Matrix

$$\Sigma = \begin{bmatrix} \text{Var}(X) & \text{Cov}(X, Y) & \text{Cov}(X, Z) \\ \text{Cov}(X, Y) & \text{Var}(Y) & \text{Cov}(Y, Z) \\ \text{Cov}(X, Z) & \text{Cov}(Y, Z) & \text{Var}(Z) \end{bmatrix}$$

Eigenvalue Decomposition

3 Eigenvectors + 3 Eigenvalues

t-Distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE : Student t Distributed Stochastic Neighbor Embedding

- Advanced version of “Stochastic Neighbor Embedding (SNE)” (Geoffrey Hinton et. al., 2002)
 - Deriving optimal reduction results by updating the mapping of the reduced dimension with gradient descent
 - **Most commonly used** as a dimensionality reduction method **for data visualization**
(specialized in **identifying clusters for dataset**)

Linear

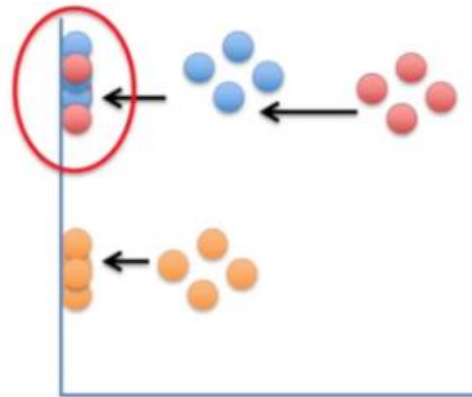
▷ Principal Component Analysis (1901)

Non-Linear

- ▷ Multidimensional Scaling (1964)
- ▷ Sammon Mapping (1969)
- ▷ IsoMap (2000)
- ▷ Locally Linear Embedding (2000)
- ▷ Stochastic Neighbor Embedding (2002)

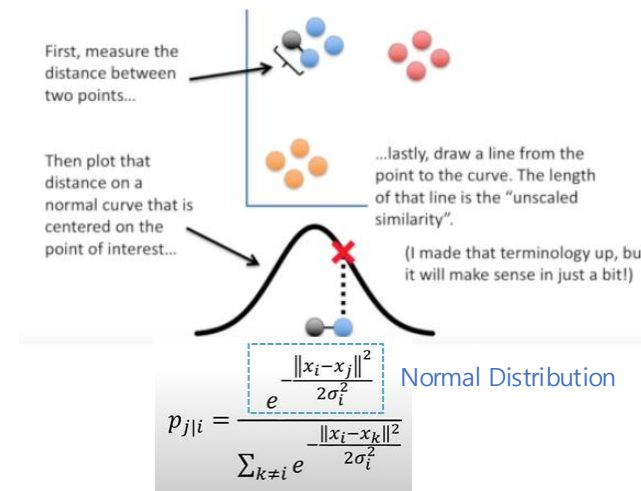
Representative Dimensionality Reduction Methodologies (Unsupervised)

* **PCA** : focusing on data variance
(data clusters are ignored)



* **SNE** : optimizing the distance (similarity)

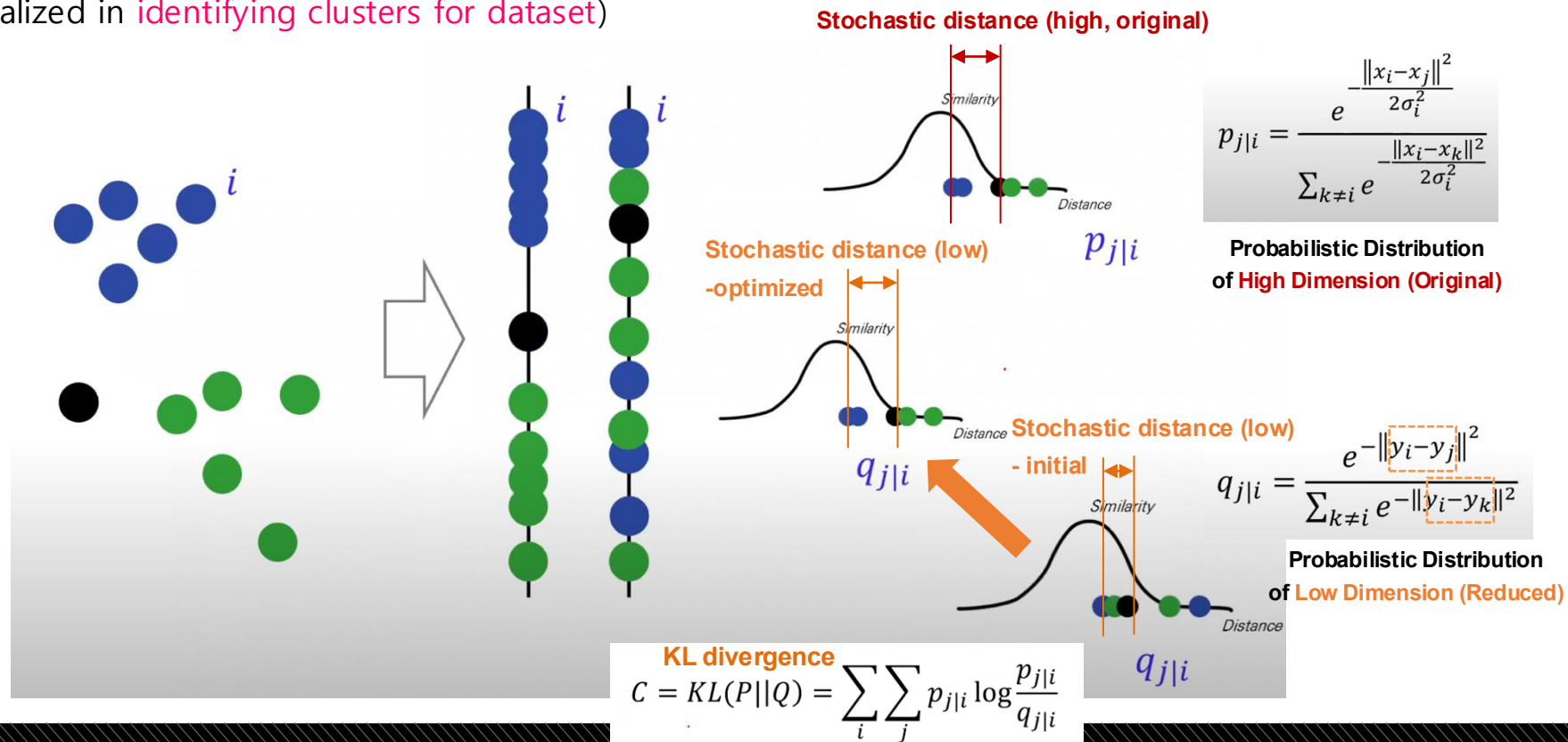
between each data sample in high & low dim.
to be similar to each other



t-Distributed Stochastic Neighbor Embedding (t-SNE)

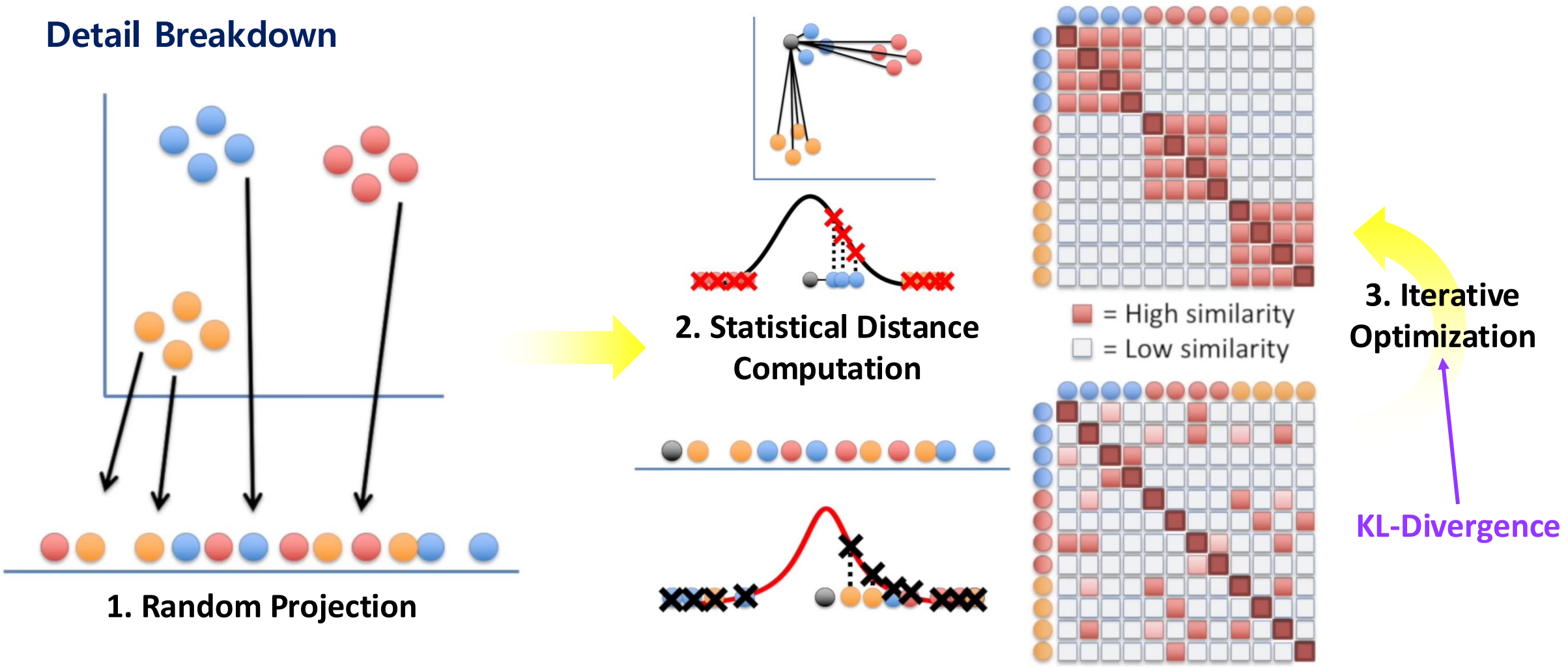
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t-Distributed Stochastic Neighbor Embedding (t-SNE)

Detail Breakdown



t-Distributed Stochastic Neighbor Embedding (t-SNE)

t-Distributed???

